# MACHINE TOOLS SYSTEMS MEM421

## HND II MANUFACTURING OPTION LECTURE Note

AUTO-MATION:- The word automation is derived from the Greek and it means self-moving or self thinking. The word automation has been used to refer to a type of manufacturing system in which the various manufacturing functions such as material processing, material/part handling and inspection are carried out by self-operating machines without human intervention.

Automation has been developed out of the need for higher productivity, lower cost and more precise manufacturing, avoidance of human intervention, omission of conventional tooling and fixturing and quick change capability of these elements are the primary factors, which were considered for automation. Hence, there ~~are~~ is an optimization of cutting tool life and quality of jobs, possibility of making parts which are impossible in conventional machining and quick and more accurate inspection and detection of error in design and fabrication.

AUTOMATIC: This means the capability or ability of operating without external control or intervention.

AUTOMATIC MACHINES: These are machines in which both the workpiece handling and the metal cutting operations are performed automatically. P.T.O

In these Machines operations, right from feeding of the Stock to clamping Machining and even inspection of the workpiece are carried out automatically.

There are Mass Production Machine tools, which are highly specialized but inflexible because of their longer setup times while changing over to new jobs as these Machines need cams, templates, stops, electrical trip dogs etc. Also there are general purpose Machine tools which are highly flexible, but are not suitable for mass production. Thus a great need of automation was felt which could bridge the gab between the highly specialized Machine tools and the general purpose Machine tools.

Hence, Automation is done incase of following cases:

1. Loading and Unloading of Parts.
2. Automatic production lines
3. Automatic tool changing
4. FMS
5. Factory automation
6. The emphasis on reducing Cycle time
7. Reduction of idle time
8. Complex parts
9. Frequently changing design.

The aim of any Manufacturing Systems is to Manufacture a Product in the most economical manner possible. Whatever may be the form of automation.

## ADVANTAGES OF AUTOMATION

In general, automation results in

1. Reduction in direct labour cost due to less operator skill required.
2. High quality of product
3. Increased productivity
4. Increased shop efficiency.
5. Prevention of shut down
6. Reduced in-process inventory
7. Reduced material handling cost and time
8. Increased manufacturing control
9. Increased safety (Human safety is fully ensured).
10. Versatility
11. Greater accuracy and repeatability and hence less rejections
12. Reduction in lead time and setup time lowers the production cost.
13. High production rates as the machining conditions are optimized.
14. Better machine utilization due to reduced idle time.
15. Changes in part design can be incorporated very easily and at a low cost.
16. Lower tooling cost as expensive jigs and fixtures are not required.
17. Reduced cycle time and increased tool life.

DISADVANTAGES

1. Very high initial cost
2. High maintenance cost
3. Skilled software knowledge specialist needed.

Automation is also answer to the reduction of problems ~~of~~ such as operator fatigue, carelessness and other human frailties.

TYPES OF AUTOMATIC CONTROLS.

1. Full automatic Control: In this type of control, the human involvement is totally eliminated and the process is entirely carried out and controlled through automatic means along with a proper feed-back system.

2. SEMI OR PARTIAL AUTOMATIC CONTROL
This means, the replacement of human activities or involvement by automatic means only partial.

3. MANUAL CONTROL: This means physical work done by hand or using basic implements instead of by machines. Usually implying it is unskilled or physical demanding

There is difference in the terms automation and Mechanization.
Mechanization means the replacement of manual labour with machines eg The movement of a tool slide with the help of a cam. However, there is no provision of feed-back, so this is an open-loop system. On the other hand, automation means a closed-loop control system in which there is provision ~~for~~ of feed back.

## PRINCIPLES OF CONTROL SYSTEMS.

A Control system is a System of Integrated elements whose function is to maintain a Process Variable at a desired Value or within a desired range of Values.

## PRINCIPLES OF EFFECTIVE CONTROL.

The effectiveness of Control System is a Process of deviation from Plans, Shortfalls, their causes etc, must be detected at minimum cost and without any unforeseen consequences.

## THE BASIC ELEMENTS OF A CONTROLLED SYSTEMS

The elements of a good control systems are.

1. Feedback
2. Control must be objective
3. Prompt reporting of deviation.
4. Control should be forward-looking
5. Flexible control
6. Hierrarchical Suitability.

## PARTS OF CONTROL SYSTEM.

There are 4 basic elements of a typical Motion Control Systems, these are.

1. Controller
2. Amplifier
3. Actuators
4. Feedback

# MACHINE TOOLS.

A Machine tool is a Powered Machine used for removing chips or unwanted Materials from a workpiece. It can also be a Machine which cuts away Surplus Materials in the form of chips from the Material supplied leaving the workpiece to the required shape and size to an acceptable degree of accuracy. ~~and~~

By this definition, it all Means that Machine tools are Metal cutting Machine which are used in workshops or Industries.

The following are the Various M/c tools found in a Production or Metal cutting ~~shop~~ workshops.

1. Lathe Machine
2. Milling machine
3. Drilling machine
4. Shaping machine
5. Slotting machine
6. Grinding machine
7. Power saw Machine
8. Shearing Machine
9. Folding machine
10. lettering Machine

P.T.O

## DESIRED PROPERTIES OF MACHINE TOOLS.

1. Ability to hold workpiece and cutting tool securely.

2. Ability to displace the tool and workpiece relative to one another to produce the required workpiece size.

3. Ability to be endowed with sufficient power to enable the tool to cut the workpiece material at economic rate.

Automatic Control Systems:- These are systems where automation drives are incorporated into the machine system to control its performance and activities.

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## TYPES OF AUTOMATION SYSTEMS

Automation Systems can be classified into four (4) major types

1. Hydraulic Automation
2. Pneumatic Automation
3. Electrical Automation
4. Mechanical Automation.

1. Hydraulic Automation:

Hydraulic drives are widely used to obtain infinitely variable rates of rectilinear motion in machine tools. Mostly it refers to feed motions but in some machine tools (planers, shapers, slotters and broaching machines) main drive speeds are varied in this way.

Hydraulic drives with a power rating of 20-45 KW are economically sound. A hydraulic drive is advisable for developing high torques and pulling forces. Its cost is less as compared to electric drive of the same rating. The application of hydraulic drive in copy system (copy turning, copy milling) and flow forming lathes is also of great importance.

ADVANTAGES OF HYDRAULIC DRIVES

1. Rapid and infinitely variable adjustment obtainable during operation, for the length, speed and direction of travel of a machine tool unit.
2. Faster reverse and acceleration rates are possible, because of less inertia and cushioning effect of the fluid.
3. The drive is smooth and reverse without shock.

4. The drive stops when fluid pressure reaches a preset maximum and tool breakage is less likely than with a mechanical drive
5. Automatic protection against overloads.
6. Ability to stall against and obstruction without damage to the tool or to the machine.
7. Convenience of remote control and its automation
8. Self lubrication of the system.

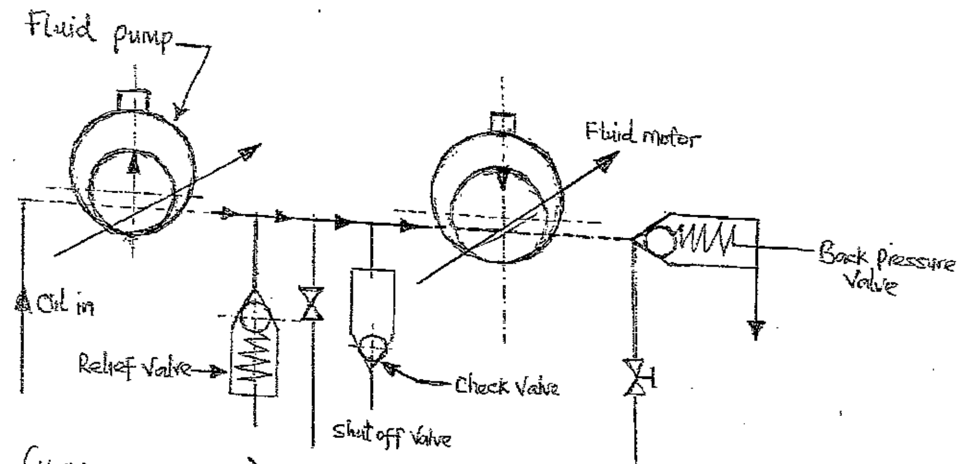
DRAWBACKS (DISADVANTAGES)

1. Insufficiently flat characteristic curve ~~result~~ resulting from leakage.
2. Effect of the temperature on the oil viscosity
3. At slow speeds (12 to 15 mm/min), the operation of the drive becomes unstable.

HYDRAULIC DRIVE FOR ROTARY MOTION.

Variable displacement pumps are inversible and can, therefore be employed either as pumps or rotary hydraulic motors. The variable capacity fluid pump with one direction of flow is run by an electric motor. The pressurised oil is used to rotate variable capacity hydraulic motor with one direction of flow. The rotary motion of the hydraulic motor is utilised to have a rotary motion of the operative unit of the machine tool. Spring relief valve limits the torque of the rotary hydraulic motor.
Check valve is operated when the rate of flow of oil circulating in the system is changed as a result of a change in the pump or hydraulic motor control factor. After one of control factors has changed, the hydraulic motor continuous for a certain time to run by ..

Inertia at the same speed but with a changed rate of flow. When this valve opens, the required amount of oil is admitted into the System, so as to compensate for the insufficiency of the circulating volume of oil. Shut off valve can be used to stop the shaft of the hydraulic motor rapidly without stopping the pump. Back pressure valve and cushioning or damping valve protect the system against shock loads in the periods of bracing and reversal.
The motors for rotary hydraulic drives are particularly complex, expensive and of low efficiency after wear and are not common.



(HYDRAULIC MOTOR)

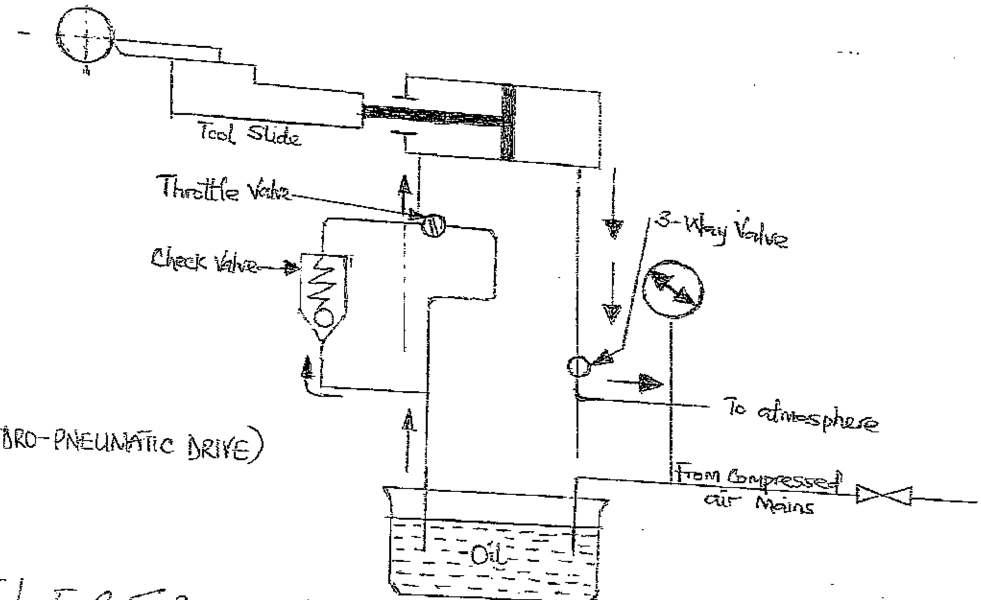
PNEUMATIC DRIVES

Many low cost automation are based on air-operated devices. Most of the manufacturing shops will have pressurized air supply for clamping, cleaning etc. This could also be used as an energy source for automation purposes. This could keep the cost of automation low and hence this type of automation is called low cost automation.

The actuators are the air motors for rotary motion and pneumatic cylinders for linear motion.

Compressed air is already available in most plant and can be put to work with rather inexpensive equipment. Air flow is fast, but the use of compressed air has several disadvantage that limit to light service. Pressures available are usually not high. A compressed air system by itself is hard to control because of the compressibility of air. Feeds and speeds are inclined to vary too much as the load changes and the equipment may not stop and reverse within desired limits.

An air-driven but hydraulically controlled circuit mitigates some of the shortcomings of air. Compressed air is admitted into the oil tank at a pressure of 0.4 to 0.5 N/mm². Consequently oil will pass through the pipe line and check valve to the left of the cylinder. This moves the piston to the left and withdraws the toolslide from the work. Air from the right end of the cylinder will be released to the atmosphere. When the tool slide is to be advanced to the work, a three way valve is turned so that compressed air is admitted to the right hand end of the cylinder and not to the oil tank. The speed of the piston, however, will be restricted by the oil draining from the cylinder as it can pass only through the throttle valve. Thus, the speed of the piston (toolslide) depends on the size of the slit in the throttle valve through which oil flows.




(HYDRO-PNEUMATIC DRIVE)

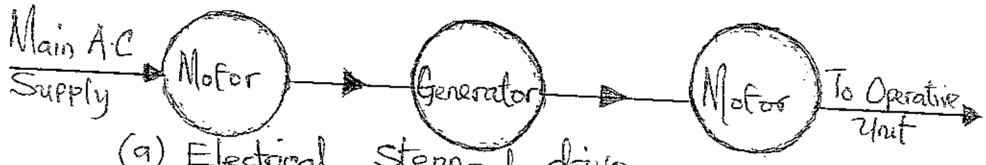
## ELECTRICAL DRIVES

The trend in the development of Machine tools drives has been towards more Complete motorisation. The electric derives of Machine tools Comprise one or several drive motors and devices for their Control. Up to date electric Motors Can be reversed or braked. The motorisation of Machine tools and an ever wider used of electric Controls will Continue to be one of the principal factors in their Improvement, in increasing the rate of production and reliability of Machine tools and in reducing operator fatique.

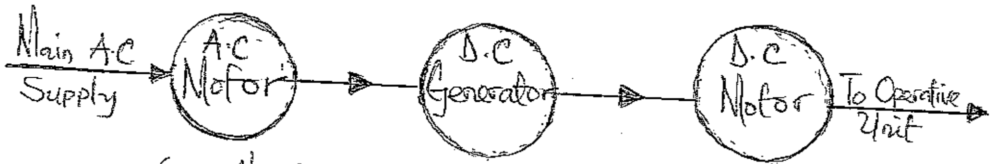
Three phase squirrel cage Motors for a power Supply of 220/380 or 500 V, 50 C/s are most frequently employed in Machine tools. Such Motors have a flat characteristics that is their speed falls only slightly with an increase of load. Three phase electric motors have speeds in rpm near to the following Series: 3000: 1500, 1000, 750, 600 and 500.

Electric motors permit momentary overloads up to 1.5 to 2 times the rated values. This feature is used in starting a motor since a considerable overload is experienced at this time.

Sig 22'28




(a) Electrical Stepped drive




(b) Ward Leonard stepless drive

(ELECTRICAL DRIVES)

# MECHANICAL DRIVES.

In Mechanical drives, the transmission elements will depend upon the type of Conversion needed between the drive shaft and the driven shaft.
The transmission elements between the input and output shafts can perform the following functions.

1. Convert rotary motion into translatory motion and Vise-Versa.
2. Convert rotary motion into rotary motion
3. Convert translatory motion to translatory

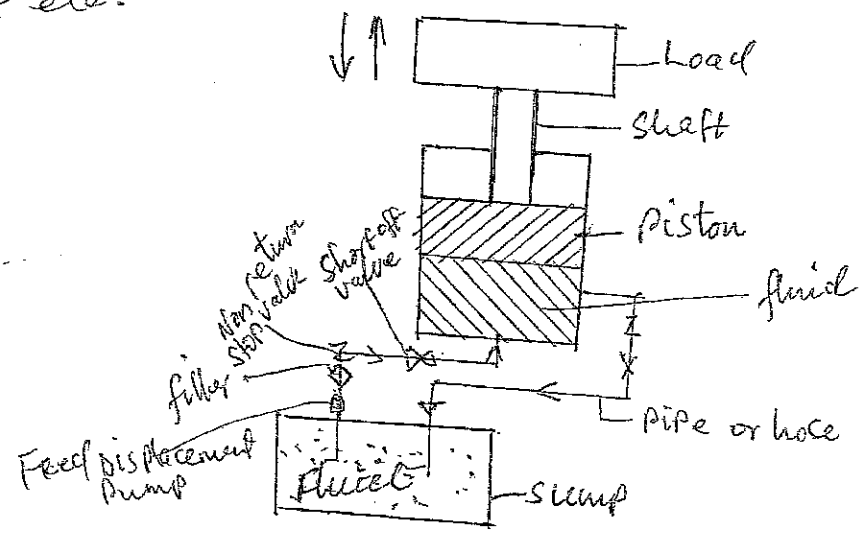
<u>Con</u>version of rotary Motion into rotary motion:- The transmission elements are used to Convert rotary motion of the drive shaft (input shaft) to the rotary Motion of the driven shaft (Output shaft)
eg ① Belt drive ② Chain drive ③ Toothed gearing.

Conversion of Rotary motion into Rectilinear reciprocating Motion. The following kinematic links are used to convert rotary motion into rectilinear Motion and are Vice-Verse.
① Rack and Pinion ② Worm and Rack
③ Screw and Nut ④ Slider and Crank.

5. Crank and slotted rocker arm
6. Cams

## HYDRAULIC SYSTEM:

Hydraulic system or hydraulic machine is the system or machine which used fluid as the medium of transmitting power. Hydraulic operated machines are used to perform series of operations. In hydraulic system, power is transmitted through the action of fluid flow under pressure. Hydraulic machine or system operates on a complete circle type of system. With this type of system, the fluid flows out of the reservoir with the aid of pump which impact pressure on the ~~liquid~~ fluid and later on returns back to the reservoir to complete the circle at the end of the operation. The process repeated itself as far as the system is in operation. Common hydraulic control application are power steering and brakes in automobiles, the hydraulic jack, press, lift etc.

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## HYDROSTATICS:

Hydrostatics is a branch of physics that deals with the characteristics of fluids at rest and especially with the pressure in a fluid or exerted by a fluid on an immersed body.

## WHAT MAKES SOMETHING HYDROSTATICS?

Hydrostatics pressure occurs when gravity pushes stagnant-dense (static) water (hydro) against below-grade walls built partially or entirely below the water table or on a hillside.

## HYDROSTATICS IN CHEMISTRY.

In this context, hydrostatics is the branch of fluid mechanics that deals with the equillibrium condition of a floating body and immersed body in the fluid at rest position. In this field, the pressure force exerted by the static fluid on the body which was immersed at a depth from the free surface of the fluid has been studied

## HYDROSTATIC PRESSURE:

Hydrostatic pressure is defined as the pressure exerted by a fluid at equillibrium at any point of time due to the force of gravity.

At this point, hydrostatics pressure is proportional to the depth measured from the surface as the weight of fluid increased when a downward force is applied.

EXAMPLE of hydrostatics pressure.

A hydrostatic pressure can be a water tower with an elevated tank, that is via pumps, filled with water.

HYDRODYNAMICS:

This is a branch of physics that deals with the motion of fluid and the forces acting on solid bodies immersed in fluids and in motion relative to them.

LAW OF HYDRODYNAMICS

It states that the fundamental principles governing hydrodynamics are the laws of conservation of mass, momentum and energy.

CH3

# NUMERICAL CONTROL MACHINE TOOL.

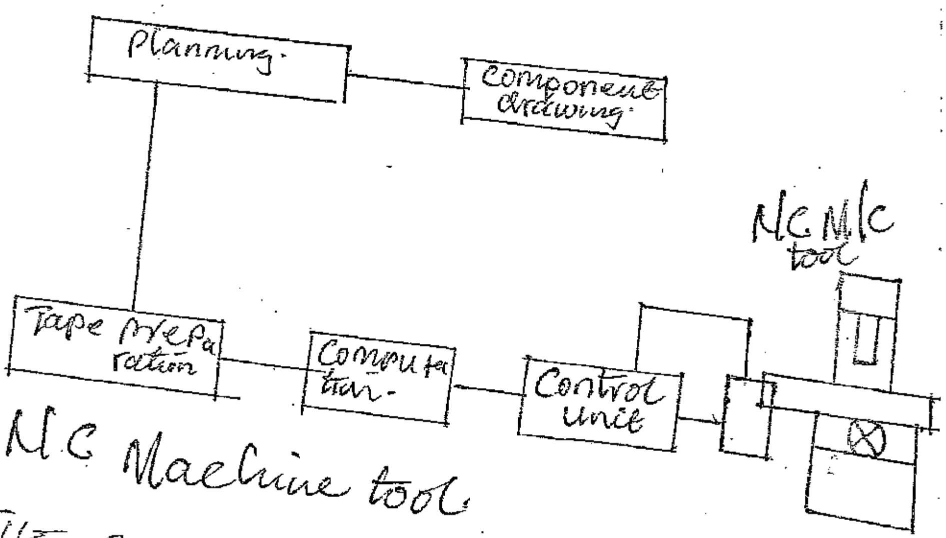
## INTRODUCTION:-

In Conventional or manually operated Machine tools, the process starts from the part drawing, and the machinist is responsible for the entire job. The machinist determines the machining strategy, sets up the machine, selects proper tooling, chooses machining feeds and speeds, and manipulates machine controls to cut a part that will pass inspection. It is clear that using this method of machining involves a considerable number of decisions that influence the accuracy and surface finish of the machined part.

Numerical Control (NC) is the operation of machine tools and certain other processing machines by a series of coded instructions, which consist mainly of digits or numbers and other symbols, including letters of the alphabet.

In the context of this basic definition, as related to metal cutting machine tools, NC commands may range from positioning of the spindle in relation to the workpiece, which is the most important function, to auxillary operations such as selecting a tool station on a turret or controlling the speed and direction of spindle rotation NC commands may even include the ON/OFF control of coolant flow.

Numerical Control (NC) is a system that uses prerecorded informations prepared from numerical data to control the machine tool or the machining process. NC describes the control of machine movements and various other functions by instructions expressed as a series of numbers and initiated via electronic control system.




NC Machine tool.

THE BENEFITS OF NUMERICAL CONTROL MACHINE TOOL.

The ultimate benefit is achieved when machining small and medium size runs. Generally, NC can be used when:-

1. The tooling cost is high compared to the machining cost by conventional method.

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2. The setup time is large in conventional machining.
3. Frequent changes in tooling and ~~M/~~ M/C setting are required.
4. Parts are produced intermittently.
5. Complex-shaped components are needed.
6. Expensive parts where human errors are costly.
7. Design changes are frequent.
8. 100% inspection is required.

ADVANTAGES OF ~~A.C~~ (N.C)

1. Greater flexibility: with NC a wide variety of operations can be performed, changeovers from one run to another through tape or program changes can be made rapidly and design changes to parts can be made rapidly through minor changes to the part program.

2. Elimination of template, models jigs and fixtures. The NC controls tape takes over the job of locating the cutting tools, which eliminate the design, manufacture, and the use of templates, jigs and fixtures.
3. Easier setups. By using more simple work holding and locating devices the operator does not have to set table limit stops or dogs or depend on the feed screw dials when setting up for machining.

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4. Reduced Machining time - Machining with NC allows the use of a wider range of speeds and feeds than conventional M/c tools. Optimum selection of feed rates and cutting speeds is ensured. The NC equipment can also move from one cutting operation to the next faster than the operator which significantly reduces the total M/cing time.

(5) Great accuracy and Uniformity: During NC Machining, no human errors are possible and machining of the same part is performed in the same way through the stored tape or program, which improves the uniformity and interchangeability of the machined parts. Therefore, inspection time is greatly reduced and is necessary for the first piece only, in addition to random checks for critical dimensions.

(6) Greate safety:- The operator is not as closely involved with the actual Machining operations as with conventional M/c tools. As the tape is checked out before actual production runs, there is less chance of machine damage that may cause human injuries.

7. Conversion to the metric system: An NC system can be converted to accept either inch or metric inputs.

# DISADVANTAGES OF NC.

1. NC follows programmed instructions that can lead to machine distruction if not properly prepared.

2. NC cannot add any extra machining capability to the machine tool, as no more power from the original drive motor and no more table travel than originally build in to the machine tool can be added.

3. NC machines cost five to ten times more than conventional machines of the same working capacity. The machine, therefore, cannot remain idle and needs special maintenance.

4. The skills required to operate an NC are usually high, because of the sophisticated technology involved, which requires part programmers, tool setters, punch operators and maintenance staff who are more educated and well-trained than conventional machine operators.

5. Special training for personnel in software and hardware is very important for successful adoption and growth of the NC technology.

6. NC requires high investment in terms of wages of highly skilled personnel and expensive spare parts.

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ADOPTION OF NUMERICAL CONTROLLED MACHINE TOOL.

Some of the main reasons for adopting Numerical Controlled Manufacturing are:-

1. NC provides flexible automation adaptable to many different requirements and changeovers from one job setup to another are rapid.

2. NC places machine operation and production directly in the hand of management through the predetermined punched tape or other media.

3. NC makes short and medium production runs economical; it can often be used economically even for certain single pieces. Therefore, high volume or mass production quantities are not always required for its economic application.

4. NC has additional advantages that may be paramount to an individual plant considering NC manufacturing reducing tooling cost in machining operations, lowering inspection and assembly costs due to improved uniformity of product, and greatly increasing machine utilization time.

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BINARY AND DIGITARY CODING SYSTEM

Binary numbers are base two (2) numbers while digitary were refers to as the normal system of numbers used in every day life. In the decimal system which uses base ten (10) as its base. In NC machines and computers, the binary system is used. This controls movement along X, Y, Z axes. The binary coding system is preferable to decimal or digitary because the binary system agrees with electrical functions. Number in the binary coding are 1 and 0 only which agrees with the principles of "ON" and "OFF" in electrical systems. For instance, 0 means 'OFF' and 1 means 'ON'. The basic industrial application is in the displacement of machine slide eg the beds and spindle rotation along X, Y, Z axis. Thus, system permits numbers to control signals like rotating the spindle, feeding the tool, work etc.

CONVERSION OF DENARY TO BINARY

To convert a number in base ten (10) to a number in base 2, we adopt the method of continous (long) division and afterwards, write the remainders from the bottom vertically upward.

P.T.O

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Example: Convert $57_{10}$ to a number in base 2.

Soln

$$\begin{array}{r|l} 2 & 57 \\ 2 & 28 \text{ r } 1 \\ 2 & 14 \text{ r } 0 \\ 2 & 7 \text{ r } 0 \\ 2 & 3 \text{ r } 1 \\ 2 & 1 \text{ r } 1 \\ & 0 \text{ r } 1 \end{array} \uparrow \quad 111001_2$$

$\therefore 57_{10} = 111001_2$

Conversion of Binary to Denary.

To Convert a number in base 2 to a number in base 10 we adopt the Method of algebraic expression assuming their powers in ascending order of magnitude from left to right.

Example. Convert $111001_2$ to a number in base 10.

Soln

$$\overset{5}{1}\overset{4}{1}\overset{3}{1}\overset{2}{0}\overset{1}{0}\overset{0}{1}_2$$

$$\therefore 1\times2^5+1\times2^4+1\times2^3+0\times2^2+0\times2^1+1\times2^0$$

$$32+16+8+0+0+1$$

$$57_{10}$$

$$\therefore 111001_2 = 57_{10}$$

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# Coding of Information in NC Machine

NC as described earlier is control by numbers, NC is control by information contained in a part program, which is a set of coded instructions given as numbers for the automatic control of a M/C in a pre-determined sequence. eg N005, G01, U20 W50 S1200 F0.2 M08;

The program block codes the information necessary to operate the lathe and is given below

N005 Block number
G01 Linear Interpolation
U20 X Increment in slide movement
W-50 Z Increment in slide movement.
S1200 Spindle speed 1200 rpm.
F0.2 Feed 0.2 mm/rev.
M08 Coolant on
; End of block.

Each of the above consist of an alphabet or a word address (N, G, U, W, S etc) and a ~~number~~ numeric value (005, 01, 20-50, 1200) which represent a function or a slide displacement position or M/cing data. PTO

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The alphanumetric data is digitally coded either in the ISO International Standardization organization) System of EIA Electronics Industries Association USA System.

## Coding of Information

Early N C Systems coded only numeric data and they did not use any alphabetic characters since all the information necessary to carry out the allowing operation is passed on to the control system through these coded numerical data, the control came to be known as numerical control (N C).

The numeric data is usually represented in decimal format. For the purpose of the convenience in processing in a digital computer, the data is to be converted (coded) into binary form. A decimal number is represented in binary form as a power of 2

To represent large number a binary coded decimal (B C D) System is employed.

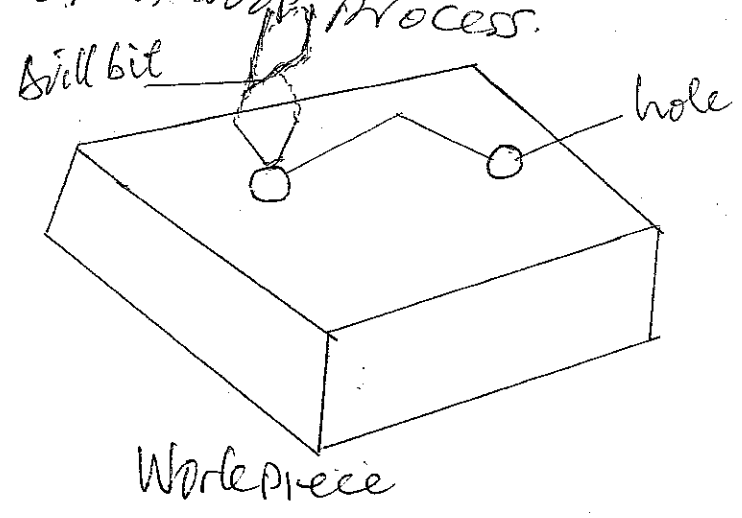
Consider the 4-track coding system shown below. Rightmost track is $2^0$ (least significant) and leftmost track is $2^3$ (most significant) for example the number 358 can be represented as

## APPLICATION OF BINARY NUMBER.

The basic principle ~~is~~ based on the displacement of machine tool slides eg Beds and rotation of the spindle along the X, Y, Z axes. This system permits numbers to verbal signals like rotational spindle, feeding the tool workpiece, Stop - NO and Proceed - Yes.

## THE PRINCIPLES AND APPLICATION OF POINT-TO POINT POSITIONING CONTROL.

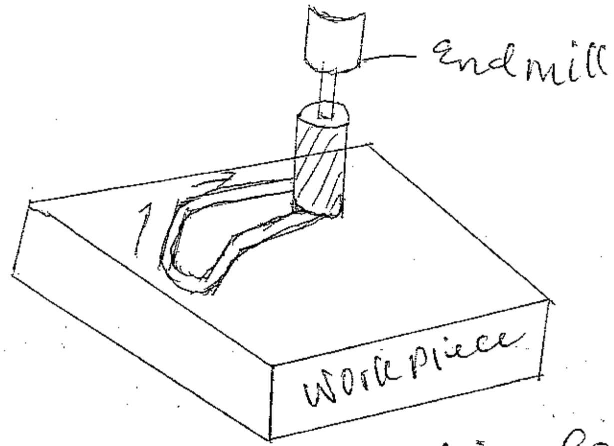
In this, the tool path is not controlled. This is usually employed in drilling of holes and welding especially spot welding. This is ~~and~~ an open-loop process.




## Straight-line NC machines.

These permit axially parallel traverse at desired table feed rate, but only one axis drive is operated at a line. This is usually found in milling machines and lathes. A continuous straight cut is possible at an angle of 90° or less PTO

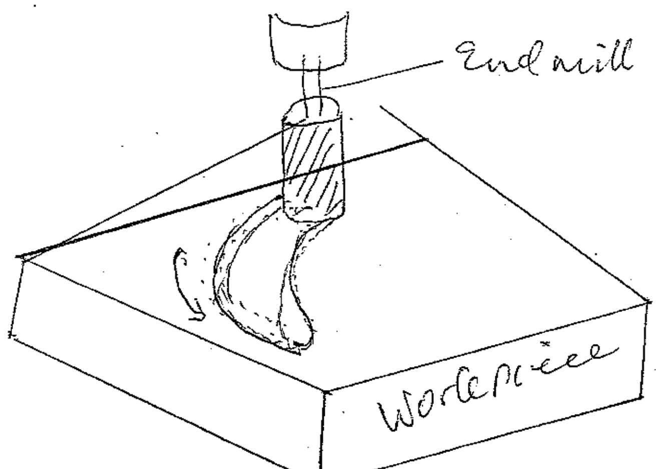
depending on the workpiece positioning.
This is a close loop process.




Straight line Control

## Contouring NC Machine.

These permit two or three dimensional geometries to be generated. It is ~~Sometimes~~ Sometime called Continuous path. It is found on Machining centres




Contouring (Continous) Path Control

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NUMERICAL CONTROL SYSTEM.

This utilizes Codes in the form of number and alphabets in providing the control of the activities required during all processes.

(a) Hydraulic Control System.

This Control System utilizes fluids under very high pressure in Controlling the activities and Motion of the process.

(b) Pneumatic Control System.

The System utilizes Compressed air to control Motion and activities during processes.

(c) Electrical Control System.

This Control System Makes use of electrical devices like, Switches, thermostat, Motors etc to Control the motion and activities of the Machine processes.

FUNCTIONS TO BE PERFORMED ON MACHINE TOOL IN ORDER TO MACHINE A PART.

1. Plan the operation sequence-
2. Select tool
3. Set and change the tool
4. Select the feeds and speeds
5. Set the feeds and speeds.
6. Position the work relative to the tool
7. Start and stop the machine operation
8. Control the relative motion (path) between the work and the tool during cutting
9. Influence motions of the work and tool to restart operation.

C44.

# TOOL LAYOUT FOR AUTOMATICS

AUTOMATIC:- This means the ability or capability of operating a machine(s) without external control or human intervention.

Automatics can also be classified according to purpose and arrangement of the spindle.

According to purpose, automatics are classified into general and single purpose machines.

According to the arrangement of the spindle automatics are classified into horizontal and vertical machines.

AUTOMATIC MACHINES: These are machines in which both the workpiece handling and the metal cutting operations are performed automatically.

## AUTOMATIC LATHES.

Automatic lathes are machine tool in which the component are machined automatically. The working cycle is fully automatic that is ~~working~~ repeated to produce duplicate parts without participation of the operator. All the working and idle operations are performed in a definite sequence by the control system adopted in the automatic which is set up to suit a given work. Strictly speaking, the machine is not fully automatic, since operator is required to load the machine for a batch of parts and start each cycle.

These machines are used when production requirement are too high for turret lathes to produce economically.

P.T.O

Their uses results in the following Advantages.

1. Greater Production over a given Period.
2. More economy in floor space
3. More Consistently accurate work than turret
4. More Consistent flow of production
5. Scrap loss is reduced by eliminating operator's error
6. During the automatic Machining operation, the operator is free to operate another Machine or inspect the completed parts.

SEMI AUTOMATICS:

Semi-automatic Machines are usually turning Machines adopted to chuck work. In these Machines, although the movements of workpiece or tool are automatically Controlled, but the workpiece has to be loaded into and removed from the chuck at the beginning and end of each cycle of operations. The Machining Cycle is automated, but the direct participation of the operator is required to start each sub-sequence Cycle, that is to Machine each subsequence workpiece. The operator loads the blank into the Machine, start the Machine, checks the work size and removes the Complete Component by hand. This Machine can be Converted into a fully Mechanized one if loading, Clamping and Unloading operations are Mechanised and also Magazine loading of workpiece is arranged.

Automatic and semi-automatic are chiefly designed to perform the following Machining

Operations. Centering, Cylindrical turning, tapered and formed surfaces, drilling, boring, reaming, spotfacing, knurling, cutting threads, facing, milling, broaching, grinding and cut-off. Additional operations such as slotting and cross drilling etc can also be performed on these machines with the help of special attachments.

CLASSIFICATION OF AUTOMATIC MACHINES:
These are various ways of classifying the automatic lathes depending upon the type of work machined. These machines are classified as

1. Magazine Loaded Automatics: These machines are for producing components from accurate separate blanks. These machines are also called automatic chucking machines.

2. Automatic Bar Machines: - Automatic bar machines are designed for machining components from bar or pipe stock. The machines are chiefly used for the manufacture of high quality fasteners (screws, nuts, and studs), bushing, shafts, rings, rollers, handles and other parts usually made of bar or pipe stock and recently of separate blanks as well.

SETTING UP OF AUTOMATICS AND SEMI-AUTOMATICS.

Setting up any machine tool involves all preparations required for manufacturing a ~~components~~ component in accordance with the accepted manufacturing process, to the specified accuracy and at a maximum rate of production.

Before setting up, it is necessary to do the following.

i. Selecting the type of blank to be used.
ii. Planning the manufacturing process.
iii. Drawing up a tooling layout in accordance with the manufacturing process.
iv. Calculating the time for each machine movement within the operating cycle.
v. Selecting the standard cutting tools, gauges and cams in accordance with the tooling layout.
vi. Designing and manufacturing all special tools.

When the above preparatory work has been completed, the set up man proceeds to set up the machine tool. The machining process is broken into separate operation elements, so that the work is completed at a maximum rate of production and at minimum cost.

## TIME REQUIRED FOR EACH OPERATION ELEMENTS

The number of spindle revolutions for each operation element is obtained by the ~~fol~~ relation

$$n = \frac{L}{f}$$

Where L is the tool travel in mm and f is the rate ~~of~~ in feed in mm per revolution.

The time required for each operation elements is determined by the formula

$$T_m = \frac{L}{fN}, \text{ min.}$$

The total time is based on the total number of spindle revolutions to complete a piece ~~if~~ if the spindle runs continuously at one speed. If different spindle speeds are used for different operation elements, it is necessary to recalculate the number of spindle revolutions in each operation element to a single basis, by using the so called equivalent revolution factor which is the ratio of the higher to the lower speed.

## CAM DESIGN

For laying out the cam, the radii of the plate cam curves are determined in the following manner.

1. The main data on the cam and the drive of the operative unit are taken from the service manual. This include
   (a) Maximum radius of cam $R_{max}$
   (b) Minimum radius of cam $R_{min}$.
   (c) height of the cam lever fulcrum above the cam centre.



P.T.O

(1) Ratio of the lever system.

(2) The cam radii are determined for each operative element from the final position of the operative unit (turret or cross-slide)

(3) For hexagonal turret slide or end working slide, $R_{max}$ corresponds to the minimum distance of the slide from the spindle nose. For cross-slide, $R_{max}$ will correspond to the position where the tool has advanced slightly beyond the spindle axis.

(4) The other radii for the final position of the operative units are determined as follows for first operation $R_1 = R_{max} - (L_1 - L_{min})$ where $L_1$ is the distance between the turret face and the spindle for first operation and $L_{min}$ is the minimum distance between the turret face and spindle.

(5) The cam radii of the beginning of the working travel for each operation is determined by subtracting the travel of the cutting tool from the radius at the end of each corresponding travel.

After knowing the various cam radii, the cam can be drawn. In doing so, the following points should be taken care of:

(i) The movement of the operative unit (turret, cross-slides) usually consists of rapid approach, working travel and rapid return.

(ii) Rapid approach and rapid return being idle motions, should be performed with minimum time loss. Therefore, the cam curve should rise or drop sharply at these places. Curves for these two movements are drawn to a special template in accordance with cycle time.

(iii). The working travel curve should provide for motion at a constant speed, that is, at a uniform rate of feed. This uniform curve is called Archimedean spiral. This curve can be draw to a special template or geometrically.

The cam is drawn full size and its surface is divided into hundredths. The lines of the hundredths are arcs drawn with a radius equal to that of the cam roll lever. The beginning and ends of working travel lobes are marked off on the corresponding hundredths, by drawing arcs of the required radii. Then the cam roll is drawn full size, tangent to the above points.

The distance between the two roll positions, that is, the cam ~~plate~~ rise is divided into as many equal parts as there are hundredths in the given working travel. The points of intersection of the arcs with Arabic numerals and Roman numerials are marked. A ~~uniform~~ curve passing through these points will ensure a uniform feed of the operative unit.

Example on how to calculate cycle time for the production of a component from a given data.

QUESTION:

A Cylinder bar of 100 mm diameter and 576 mm length are turned in a single pass operation. The spindle speed used is 144 r.p.m and the total feed is 0.2 mm/rev. Taylor's tool life relationship is $VT^{0.75} = 75$ where V is the cutting speed (m/min) and T is the tool life (min). Calculate

(i) Time for turning one piece

(ii) The average total change time per piece given that it takes 3 minutes to change the tool each time.

(iii) The time required to produce one piece given that the handling time is 4 min.

Soln

(i) Using the formula

$$T_m = \frac{L}{fN}$$

$$T_m = \frac{576}{0.2 \times 144}$$

$$= \frac{576}{28.8}$$

$$= 20\,min$$

Given data
$L = 576$
$f = 0.2$
$N = 144$

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P.T.O

(ii) ~~Taylor equation~~ Using Taylor's equation

$VT^{0.\cancel{78}} = 75$

Now $V = \frac{\pi D N}{1000} \Rightarrow \frac{\pi \times \cancel{100} \times 144}{10\cancel{00}}$

$= \frac{3.142 \times 1 \times 144}{10}$

$= \frac{452.448}{10}$

$45.24$ m/min.

From here $T = 1.967$ min

∴ Number of tool change during the cutting of one piece $= \frac{20}{1.967}$

$= 10$ times

Total tool change time $= 10 \times 3 = 30$ min

Total time to machine one job $20 + 30 = 50$ min

Now handling time $= 4$ min

Total time needed to produce one piece

$= 50 + 4$

$= 54$ min

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Example 2

A 15mm diameter HSS drill is used at a cutting speed of 20m/min and a feed rate of 0.2 mm/rev. Under these conditions the drill life is 100 min. The drilling length of each hole is 45mm and the time taken for idle motion is 20s. The tool change time is 300s. Calculate

(a) Number of holes produced using one drill

(b) Average production time per hole.

Soln

Using the formula $V = \frac{\pi D N}{1000}$

Making N the subject formula we have

$$N = \frac{1000 \times V}{\pi D}$$

$$N = \frac{1000 \times 20}{3.142 \times 15}$$

$$= \frac{20{,}000}{47.13}$$

$$= 424.36 \text{ rev/min}$$

∴ Using the formula $T_m = \frac{L}{fN}$

$$\frac{45 \times 15\pi}{0.2 \times 1000 \times 20}$$

$$T_m = 0.53 \text{ min}$$

$$\therefore 0.53 \times 60$$

$$= 31.8 \text{ seconds.}$$

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(ii) ~~Taylor equation~~ Using Taylor's equation

$VT^{0.73} = 75$

Now $V = \frac{\pi D N}{1000} \Rightarrow \frac{\pi \times 100 \times 144}{1000}$

$= \frac{3.142 \times 1 \times 144}{10}$

$= \frac{452.448}{10}$

$45.24 \text{ m/min}$.

From here $T = 1.967 \text{ min}$

∴ Number of tool change during the cutting of one piece $= \frac{20}{1.967}$

$= 10 \text{ times}$

Total tool change time $= 10 \times 3 = 30 \text{ min}$

Total time to Machine one Job $20 + 30 = 50 \text{ min}$

Now handling time $= 4 \text{ min}$

Total time needed to produce one piece

$= 50 + 4$

$= 54 \text{ min}$

(a) Number of holes produced using one drill

$$\frac{300 \text{ sec}}{31.8 \text{ sec}} = 9.43$$

say 9 holes.

(b) Average production time per hole =

Machining time + Idle motion time + tool change time

$= 31.8 + 20 + \frac{300}{9}$

$= 31.8 + 20 + 33.33$

$= 85.1$ seconds.

# TRANSFER MACHINES:

A transfer Machine can be defined as a large automatic installation Consisting of Several individual Machining heads or Units which are fastened together by Conveying units. Workpiece are loaded at one end and are automatically transferred along with their fixtures, from Station to Station. At each Station, Machining operation are performed on the workpieces and the Completed workpieces leave at the other end of the Machine.
Thus, a transfer Machine is a Combined Material handling and Material Processing Machine. These are basically Special Purpose Machines and are often the most Suitable Method for a Continuous Manufacture of identical or Very Similar parts in Mass Production.
Thus, a transfer Machine is economically justified only if the Continuous production of a Workpiece is met by an equal demand for it.

The Machining Station on a transfer Machine Consist of Powerhead Production Units.
A powerhead unit Consists essentially of:-

1. Powered spindle/spindles mounted in Suitable bearings, which are usually driven from a Self-Contained Motor through of gear box.

2. A Means ~~for~~ for Power feed to traverse the head along the Slideways while the Cutting tools Work upon the components. The Power feed may be Mechanical or hydraulic.

The Method commonly used is a rapidly moving return conveyor parallel to the main transfer line. If enough floor space is not available then L, U, Square or Rectangular pattern can be adopted. In the case of square or rectangular pattern, there is no need of return conveyor, since the pallets are automatically returned to the loading point. When transfer pallets are used, they contain locating holes or points that mate with retracting pins or fingers at each machining station. The pallets are then located and then clamped in proper position. When pallets are not used, locating bosses and points are designed into the workpiece itself.

(2) ROTARY TRANSFER MACHINES.

The rotary transfer machine is used when only 6 to 10 or fewer machining stations need to be employed. The machining heads are arranged radially along the periphery of a rotary indexing table. The table rotates about a vertical axis and its movement may be continuous or intermittent. For indexing the table, a Geneva type indexing mechanism can be employed. This type of machine is very compact and there is saving in floor space. The operations performed on a transfer machine include:- (1) Drilling (2) Boring (3) Counterboring (4) Reaming (5) Tapping (6) Counter sinking (7) chamfering (8) Face milling (9) Hollow milling (10) Trepaning (11) Gauging (Air or Electrical) (2)

(12) Blow-out or dumping chips (13) Rolling over or revolving work to reposition for the following operations.

The most frequent machine tools associated with a ~~tr~~ transfer machine are:-
(1) Drilling machines (2) Reaming machines
(3) Single point boring machines (4) Milling machines
(5) Tapping machine (6) Inspection equipment.

Less frequently used machine tools include:-
(1) Broaching machines (2) Boring machines
(3) Polishing machines (4) Turning machines.

ADVANTAGES OF TRANSFER MACHINES.

1. The Machining operations are speeded up thereby reducing the Production Cycle.
2. Fewer operation are needed.
3. Secure higher output.
4. Manual handling of the Component is avoided, except loading and unloading.
5. A chip Conveyor can be used for the removal of chips
6. The alignment of the job at each Machining Station is Simplified and automized.
7. Greater accuracy of the job is achieved
8. Lower cost of Production
9. Better use of floor space.

DISADVANTAGES

1. This System is justified only for high Production of Components.
2. Initial cost is ~~high~~ very high.
3. The whole setup is to be changed if the design of the component changes.
4. The breakdown of one Machine will stop all the Machines.
5. Electrically, the system is very complex.

# TURRETS LATHES MACHINE:

Turrets Lathe Machine is a Machine tool that can perform Internal and external work such as turning, boring, drilling, facing, recessing, reaming, tapping, threading, forming, knurling, chamfering and parting off etc. The operations which are usually performed from the Hexagonal turret include:- turning, boring, drilling, reaming, tapping, and threading.

The operations which are done from cross-slides include:- Facing, forming, knurling, recessing, chamfering and parting off.

Internal cuts are almost always made by tools in hexagonal turret.

The cuts which can be taken on a turret Lathe are of three types.

1. Combined cut: A combined cut is one where tools in both the hexagonal turret and square-slide turret are made to cut at the same time.

2. Multiple cut:- A multiple cut is one where two or more tools are applied at the same time from the turret station.

3. Successive cut: A successive cut is one where cuts are made from successive faces of the turret consecutively.

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The parting off operation is always done from the rear cross-slide in the case of bridge type of cross-slide.

## ADVANTAGES OF TURRET LATHE

The turret lathe is adapted to quantity production work. For this the turret lathe has got the following advantages over a simple engine lathe.

1. A large number of tools can be permanently set up in the machine in the proper sequence of their use.
2. Each cutting position is provided with a feed stop or feed trip which ensures identical cuts on successive pieces.
3. Multiple cuts and combined cuts be taken at the same time.
4. The turret lathe can be fitted with various attachment such as for taper turning, thread chasing and duplicating. It can be also tape controlled.
5. A less skilled operator is required.

## MAIN PARTS OF TURRET LATHE

The main parts of a turret lathe are as given below:

(1) Head stock: The head stock of a turret lathe is similar in construction to that of the engine lathe. However, it is larger and heavier in construction and with a wider choice of speeds and is fitted with a more

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Most threads are cut on a turret from the hexagonal turret. Self-opening dies and collapsible taps allow rapid retraction of the tool after the thread is cut. The length of cut, however, is limited. Threading is done mostly on the workpiece. For longer threads, a single point threading tool in the front cross slide must be used. This method is much slower, when a good, clean and accurate thread is desired.

In the production of holes, provision must be made for the proper order of internal cuts, eg a roughing operation using a drill is usually performed first to remove metal quickly. Then another roughing operation is carried out with a boring bar, after which a reamer is used to size the hole quickly and give a good finish.

There are three methods to obtain taper on a turret lathe ① With a form cutter; this method is suitable only for short, steep tapers.
② With a taper turning attachment as on an engine lathe
③ With a roller-rest ~~taper~~ attachment. In this method, a taper guide bar is mounted over the work in the same position as the pilot bar, replacing it. The cutting tool resembles the bar turner. It can be easily adjusted for size by means for a graduated dial. The tool and the rolls recede as they progress along the work, guided by the bar, thus producing the desired taper.

Powerful motor. Speed may range from 30 to 2000 r.p.m. There are two principal types of head stocks ① An electric head in which a variable speed motor is mounted directly in the head stock.

② The all geared head which gives a wider range of speed and allows heavier cuts to be taken.

④ Turret:- On all but very small capstan lathes the turret is a hexagonal block mounted on vertical ~~t~~ tool holder. To fasten the tool holder in position, four set of pins are passed through the drilled holes in the flange into matching drilled and tapped holes in the turret face.

A large star wheel at the front of the machine is used for moving the turret forward for each tool position. The forward movement is controlled by a preset stop and then the turret is returned to its starting position. As it returns, the turret is automatically indexed one-sixth of a resolution, bringing the next tool in the cutting position. There are six adjustable stops, one ~~of~~ for each of the six turret faces and each stop is adjusted during the set up. These six preset stops are indexed automatically whenever the turret is indexed, so that the stop corresponding to a cutting tool is in use. The turret is moved to the workpiece by hand, and feeding may be done by hand or by power feed.

For Power feeding when the desired turret travel is reached, an automatic trip lever stops further movement of the tool by disengaging the drive clutch.

(3) Cross slide:- There are two types of cross-slides which can be used for turret lathes (i) Slide hung type (ii) reach over type.
In side hung type, the carriage does not extend across the entire top of the bed, but is supported entirely on the front way and another way located on the lower front portion of the bed.

(ii) The reach over or bridge type of cross-slide is supported on the two bed ways in the conventional manner. In addition it may also be supported by a lower rail.

INDEXING OF TURRET AND CROSS SLIDES.
In turret Lathes, the turret slide (saddle) travels directly along the bed ways. The turret slide gets its movement from a rack and pinion drive. The shaft on which the large star-wheel is mounted (the starwheel is for providing hand feed to the turret slide) carries a pinion on its inner end. This pinion engages with a rack on the slide to provide longitudinal motion in either direction. For Power feed, a shaft passes through the apron and is driven from the feed gearbox at the base of the headstock.